PROJECT REPORT

Smart Invoice Validation System

Simulated SAP Procure-to-Pay (P2P) Environment

Submitted By

Name	Harsh Raj
Roll Number	2330162
Program	B.Tech — Electronics & Computer Science (ECSc.)
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Course	SAP ABAP Development Training

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1. Problem Statement

In enterprise procurement, invoice processing errors are a leading cause of financial leakage and audit failures. Traditional manual invoice verification is time-consuming, error-prone, and does not scale with business volume. Organizations using SAP ERP routinely face three critical failure modes:

Issue	Business Impact
Vendor invoices quantities exceeding received goods	Overpayment to vendors
Invoice prices higher than agreed PO price	Undetected cost inflation
Invoices raised before Goods Receipt is posted	Payment without delivery confirmation

This project addresses all three failure modes through an automated, rule-based validation system implemented in ABAP, simulating the SAP 3-way matching process (Purchase Order → Goods Receipt → Invoice).

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2. Solution & Features

2.1 System Architecture

The system uses a custom Z-table (ZINV_VALIDATION) to simulate the SAP tables involved in a real P2P cycle. This approach represents a simulated SAP environment and mimics MM-FI integration without requiring access to production SAP tables.

SAP Table (Real)	Z-Table Simulation	Data Represented
EKKO / EKPO	ZINV_VALIDATION	Purchase Order Header & Items
MSEG	ZINV_VALIDATION	Goods Receipt Quantities
RBKP / RSEG	ZINV_VALIDATION	Invoice Header & Line Items

2.2 Core Validation Logic (3-Way Matching)

The program implements a custom validation layer before FI posting using multi-condition ABAP logic. The logic is evaluated in priority order:

```
IF gr_qty = 0             → STATUS = 'BLOCKED'   (No Goods Receipt found)
ELSEIF inv_qty > gr_qty   → STATUS = 'BLOCKED'   (Qty overbilled)
ELSEIF inv_price ≠ po_price → STATUS = 'WARNING' (Price mismatch)
ELSE                      → STATUS = 'VALID'     (OK for FI posting)
```

2.3 Program Modules

Module	Transaction	Purpose
FETCH_DATA	SELECT from ZINV_VALIDATION	Retrieves invoices based on selection screen filters

VALIDATE_DATA	Internal ABAP logic	Applies 3-way match rules; computes qty/price differences
DISPLAY_ALV	CL_SALV_TABLE	Renders results in interactive ALV grid with Excel export

2.4 Key Features

- Selection Screen: Filter by Invoice ID range, Vendor ID, PO Number, and Status
- 3-Way Matching: Validates quantity and price simultaneously across PO, GR, Invoice
- Smart Calculations: Auto-computes Quantity Difference and Price Difference per record
- Status Indicators: VALID (green), WARNING (yellow), BLOCKED (red) for quick review
- ALV Grid Output: Built using CL_SALV_TABLE with column optimization and striped rows
- Excel Download: Full export capability via ALV toolbar standard functions
- Layout Save: Users can save and restore preferred ALV layouts

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3. Technical Stack

Component	Technology / Tool
Programming Language	ABAP (Advanced Business Application Programming)
Platform	SAP NetWeaver / SAP S/4HANA (Simulated Environment)
UI Framework	CL_SALV_TABLE — Object-Oriented ALV Grid
Database Table	ZINV_VALIDATION (Custom Z-Table)
Development Tools	SE11 (Table), SE38 (Program), SE16 (Data Browser)
Data Insertion	SM30 / Custom ABAP loader program
Export Format	Excel (.xlsx) via ALV built-in function

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4. Implementation Steps

Step 1 — Database Table Creation (SE11)

A custom transparent table ZINV_VALIDATION was created in transaction SE11 with fields for INV_ID, VENDOR_ID, PO_NUMBER, PO_QTY, GR_QTY, INV_QTY, PO_PRICE, INV_PRICE, QTY_DIFF, PRICE_DIFF, and STATUS. The table was activated with Data Class APPL0 and made available for maintenance via SM30.

Step 2 — Test Data Entry (SE16 / SM30)

Five sample invoice records (INV001 to INV005) were inserted covering all validation scenarios: VALID, BLOCKED (qty overbilling), WARNING (price mismatch), BLOCKED (no GR), and VALID (partial delivery). This ensures complete branch coverage of the validation logic.

Step 3 — ABAP Program Development (SE38)

The main executable program ZINV_VALIDATION_CHECK was created in SE38. It is structured into three modular FORMs:

- FETCH_DATA — Executes SELECT query on ZINV_VALIDATION based on selection screen filters (Invoice ID range, Vendor ID, PO Number, Status)
- VALIDATE_DATA — Applies 3-way matching rules in priority order; computes QTY_DIFF and PRICE_DIFF fields dynamically
- DISPLAY_ALV — Renders output using CL_SALV_TABLE with column optimization, striped rows, and all toolbar functions enabled

Step 4 — Testing and Validation

The program was tested against all five test scenarios including valid matches, quantity overbilling, price inflation, missing goods receipts, and partial deliveries to ensure correct status assignment and error handling.

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5. ABAP Program Code

Program Name: ZINV_VALIDATION_CHECK | Transaction: SE38

```
REPORT zinv_validation_check.
TABLES: zinv_validation.

TYPES: BEGIN OF ty_invoice,
  inv_id      TYPE char10,
  vendor_id   TYPE char10,
  po_number   TYPE char10,
  po_qty      TYPE i,
  gr_qty      TYPE i,
  inv_qty     TYPE i,
  po_price    TYPE p DECIMALS 2,
  inv_price   TYPE p DECIMALS 2,
  qty_diff    TYPE i,
  price_diff  TYPE p DECIMALS 2,
  status      TYPE char10,
END OF ty_invoice.

DATA: it_invoice TYPE TABLE OF ty_invoice,
      wa_invoice TYPE ty_invoice.

DATA: lo_alv      TYPE REF TO cl_salv_table,
      lo_columns  TYPE REF TO cl_salv_columns_table,
      lo_display  TYPE REF TO cl_salv_display_settings,
      lo_functions TYPE REF TO cl_salv_functions_list,
      lx_msg      TYPE REF TO cx_salv_msg.

SELECTION-SCREEN BEGIN OF BLOCK b1 WITH FRAME TITLE TEXT-001.
  SELECT-OPTIONS: s_invid  FOR zinv_validation-inv_id,
                  s_vendor FOR zinv_validation-vendor_id,
                  s_po     FOR zinv_validation-po_number.
  PARAMETERS:     p_status TYPE char10.
SELECTION-SCREEN END OF BLOCK b1.

INITIALIZATION.
  TEXT-001 = 'Smart Invoice Validation - Search Parameters'.

START-OF-SELECTION.
  PERFORM fetch_data.
```

```
  PERFORM validate_data.
  PERFORM display_alv.

FORM fetch_data.
  SELECT inv_id vendor_id po_number po_qty gr_qty inv_qty
         po_price inv_price status
    FROM zinv_validation
    INTO CORRESPONDING FIELDS OF TABLE it_invoice
   WHERE inv_id  IN s_invid
     AND vendor_id IN s_vendor
     AND po_number IN s_po.
  IF sy-subrc <> 0.
    MESSAGE 'No invoice records found for the given criteria.'
            TYPE 'S' DISPLAY LIKE 'E'.
    LEAVE LIST-PROCESSING.
  ENDIF.
ENDFORM.

FORM validate_data.
  LOOP AT it_invoice INTO wa_invoice.
    wa_invoice-qty_diff   = wa_invoice-inv_qty - wa_invoice-gr_qty.
    wa_invoice-price_diff = wa_invoice-inv_price - wa_invoice-po_price.
    IF wa_invoice-gr_qty = 0.
      wa_invoice-status = 'BLOCKED'.
    ELSEIF wa_invoice-inv_qty > wa_invoice-gr_qty.
      wa_invoice-status = 'BLOCKED'.
    ELSEIF wa_invoice-inv_price <> wa_invoice-po_price.
      wa_invoice-status = 'WARNING'.
    ELSE.
      wa_invoice-status = 'VALID'.
    ENDIF.
    MODIFY it_invoice FROM wa_invoice.
  ENDLOOP.
ENDFORM.

FORM display_alv.
  TRY.
    cl_salv_table=>factory(
      IMPORTING r_salv_table = lo_alv
      CHANGING  t_table      = it_invoice ).
    lo_display = lo_alv->get_display_settings( ).
    lo_display->set_striped_pattern( abap_true ).
    lo_display->set_list_header( 'Smart Invoice Validation Report' ).
    lo_functions = lo_alv->get_functions( ).
    lo_functions->set_all( abap_true ).
    lo_columns = lo_alv->get_columns( ).
    lo_columns->set_optimize( abap_true ).
    lo_alv->display( ).
  CATCH cx_salv_msg INTO lx_msg.
    MESSAGE lx_msg->get_text( ) TYPE 'E'.
  ENDTRY.
ENDFORM.
```

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6. Database Table Definition — ZINV_VALIDATION

Field Name	Type	Length	Key	Description
INV_ID	CHAR	10	YES	Invoice ID (Primary Key)
VENDOR_ID	CHAR	10	NO	Vendor Identifier

PO_NUMBER	CHAR	10	NO	Purchase Order Number
PO_QTY	INT	10	NO	Purchase Order Quantity
GR_QTY	INT	10	NO	Goods Receipt Quantity
INV_QTY	INT	10	NO	Invoice Quantity
PO_PRICE	CURR	13,2	NO	Purchase Order Unit Price
INV_PRICE	CURR	13,2	NO	Invoice Unit Price
QTY_DIFF	INT	10	NO	Computed: INV_QTY - GR_QTY
PRICE_DIFF	CURR	13,2	NO	Computed: INV_PRICE - PO_PRICE
STATUS	CHAR	10	NO	VALID / WARNING / BLOCKED
CURRENCY	CUKY	5	NO	Currency Key (INR)

Technical Settings: Data Class — APPL0 | Size Category — 0 | Buffering — Not Allowed

7. Test Cases & Expected Output

Five test scenarios were designed to exercise all branches of the validation logic. Each record in the ZINV_VALIDATION table corresponds to one scenario.

Invoice	PO Qty	GR Qty	Inv Qty	PO Price	Inv Price	Status
INV001	100	100	100	₹500	₹500	**VALID**
INV002	50	30	50	₹45,000	₹45,000	**BLOCKED**
INV003	200	200	200	₹150	₹175	**WARNING**
INV004	5	0	5	₹2,00,000	₹2,00,000	**BLOCKED**
INV005	20	15	15	₹8,000	₹8,000	**VALID**

INV001 — VALID: All three quantities match and price equals PO price. Invoice is cleared for FI posting.

INV002 — BLOCKED: Invoice quantity (50) exceeds GR quantity (30). Vendor is billing for 20 units that were never received. Payment blocked.

INV003 — WARNING: Quantities match but invoice price (₹175) is higher than PO price (₹150). Excess of ₹25/unit flagged for AP review.

INV004 — BLOCKED: GR quantity is 0 — goods have not been received. Invoice cannot be processed without delivery confirmation.

INV005 — VALID: Partial delivery scenario. GR qty = 15 (partial of PO qty 20). Invoice correctly raised for 15 units only. Validated successfully.

8. Unique Points

1. Real-World Business Logic Implements genuine SAP 3-way matching — the same logic used in SAP MM-FI integration in production systems. Unlike basic CRUD projects, this validates quantity AND price across three document types simultaneously.
2. Custom Validation Layer Before FI Posting The program acts as a pre-posting gateway, mimicking how SAP blocks invoices in a real AP workflow. This demonstrates understanding of the full P2P cycle.
3. Modern OOP ALV (CL_SALV_TABLE) Uses the object-oriented ALV class instead of older function module-based ALV, reflecting current SAP development best practices.
4. Computed Difference Fields Automatically calculates Quantity Difference (INV_QTY - GR_QTY) and Price Difference (INV_PRICE - PO_PRICE) to give AP teams instant visibility into the magnitude of discrepancies.
5. Multi-Scenario Test Coverage Five distinct test cases cover all code branches: valid match, qty overbilling, price inflation, missing GR, and partial delivery — ensuring complete validation of the program logic.

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9. Future Improvements

- Real SAP Table Integration: Connect to live EKKO, EKPO, MSEG, RBKP, RSEG tables instead of the Z-table simulation, enabling deployment in a production SAP environment.
- Tolerance-Based Matching: Allow configurable price tolerance (e.g., ±2%) so minor rounding differences do not trigger unnecessary WARNING statuses.
- Workflow Integration: Route WARNING status invoices to an SAP workflow (WS-based) for AP manager approval before FI posting.
- Email Notification: Auto-send email alerts to the Accounts Payable team when BLOCKED invoices are detected, using SAP Business Workplace (SBWP).
- SAP Fiori Dashboard: Build a SAPUI5/Fiori front-end to visualize invoice validation statistics — total invoices, blocked count, warning rate — as a management dashboard.
- Audit Log: Record every validation run with timestamp and user ID in a separate log Z-table for compliance and traceability.